
Lecture 16

ChE 467/596-012
Polymer Rheology

Today's Agenda

*More on gelation and gel point
- real data*

Gel point & fractals – what is a fractal?

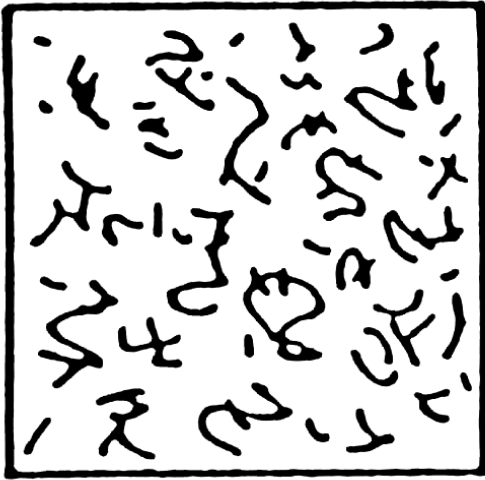
Some questions

How would you monitor gelation? What experiments?

How do you get gel point?

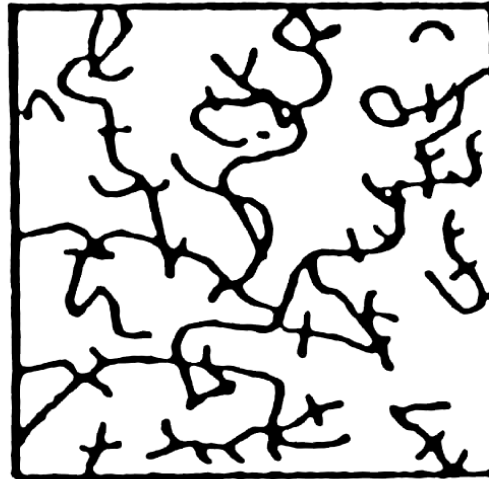
Polymer Crosslinking & Gel Point

Different stages during polymer crosslinking



Initial Stages

Clusters of molecules
Sample creeps



Gel Point

Sample-spanning cluster
formed



Final Stages

Dense polymer network
Mechanically strong

Gel point Determination

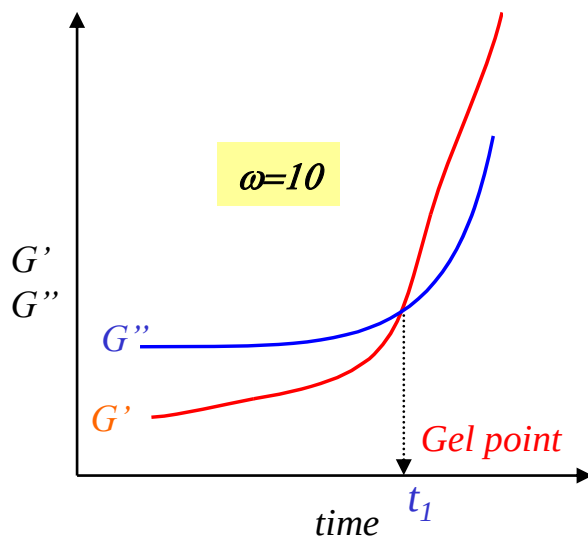
Historically-use G' and G'' cross-over

Why G' and G'' cross-over?

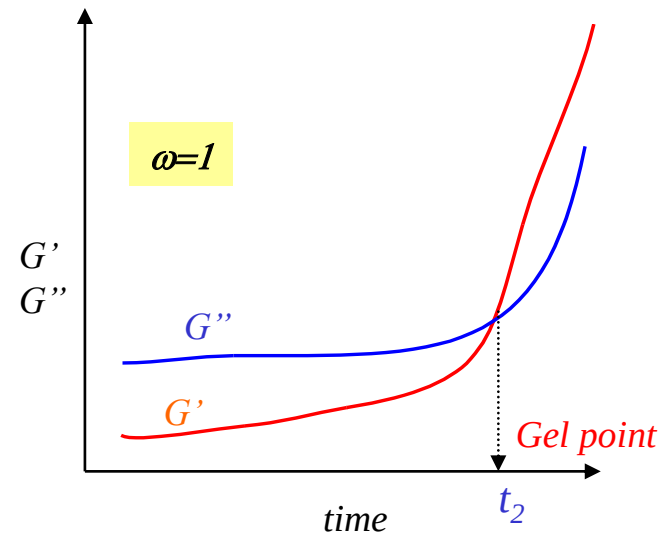
G' reflects elastic (solid) behavior; G'' reflect viscous (liquid) behavior

Gel point in between a solid and a liquid

- The technique is ok if gel point is frequency independent, if $t_1 = t_2$*



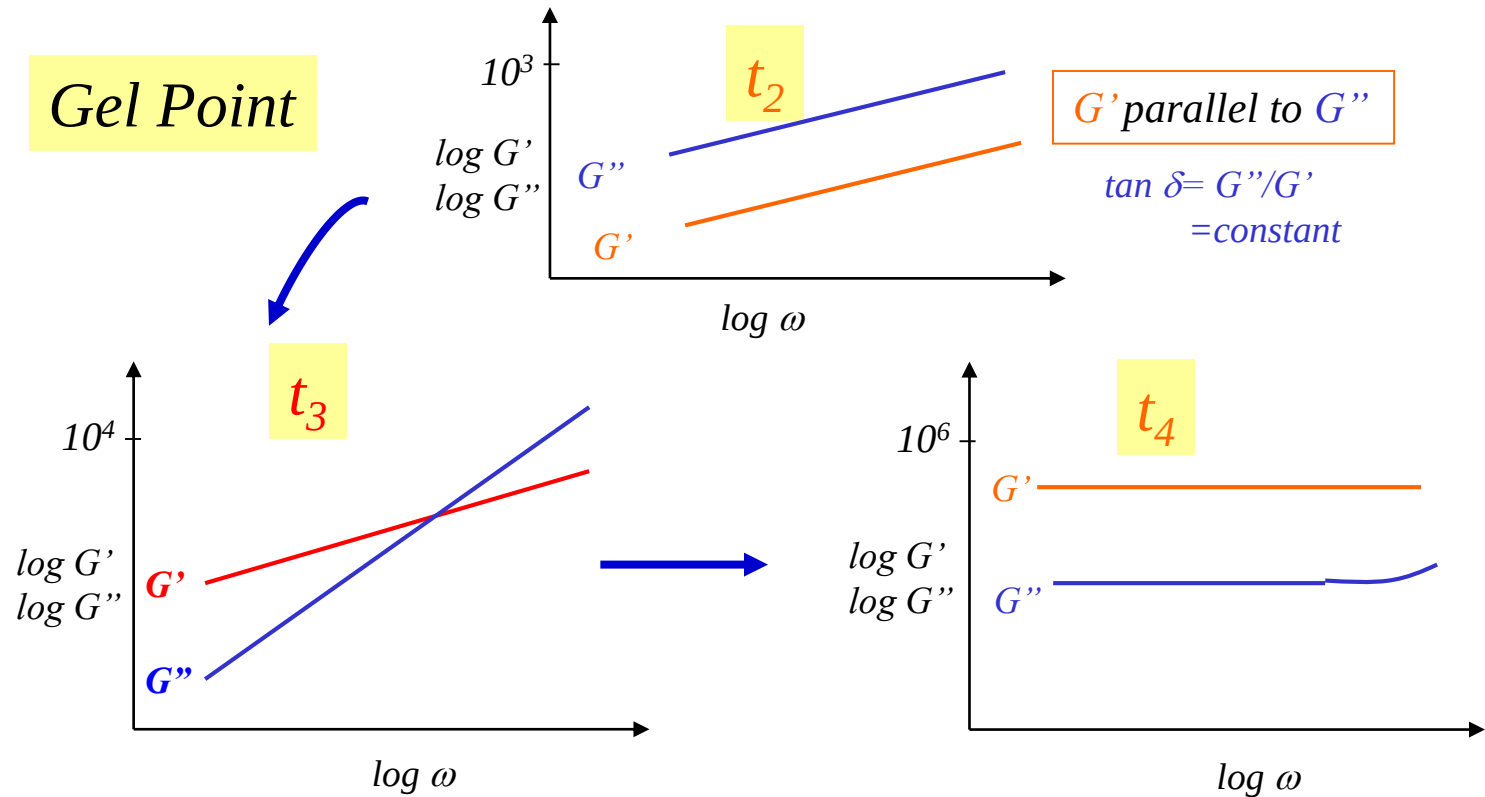
$t_1 \neq t_2$



- But often this may not be the case.*

Gelation & Gel point detection

Gel Point



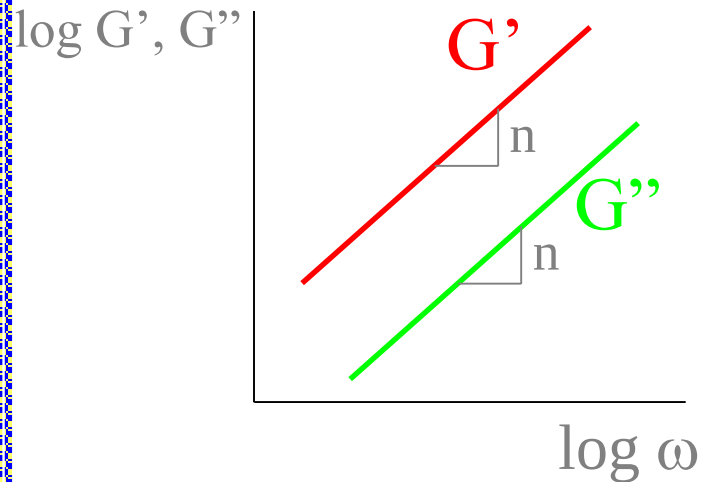
- **Note:** G' starting to crossover G''
- $G' > G''$ at low ω

$G' \gg G''$
 G', G'' frequency independent
 = **Fully x-linked gel**

Criterion for Gel Point

Winter-Chambon Criterion

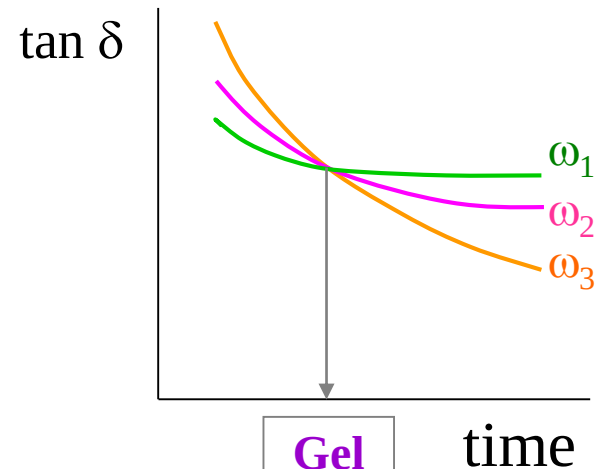
Frequency sweeps at gel point



Elastic Modulus $G' \sim \omega^n$

Viscous Modulus $G'' \sim \omega^n$

Loss tangents at different frequencies



Gel Point

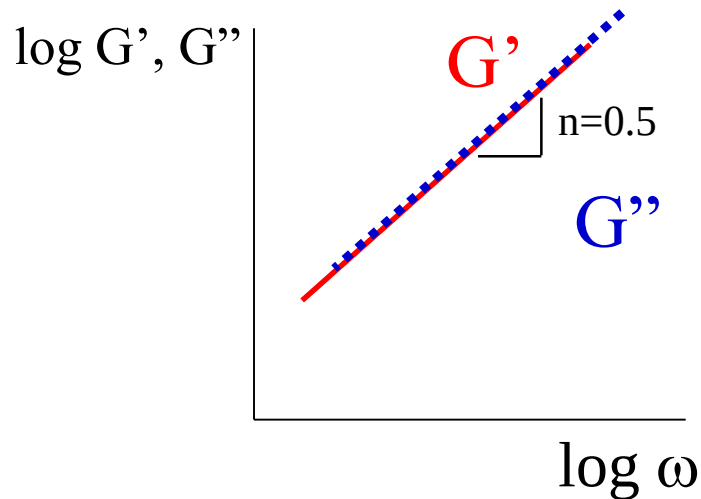
$$\tan \delta = \frac{G''}{G'} = \tan \left(\frac{n\pi}{2} \right)$$

(**independent** of frequency)

Gel Point Problem

Experiments show that G' and G'' cross-over is independent of frequency

What would the G' , G'' versus frequency look like at the gel point?
What is $\tan \delta$ in this case?



Elastic Modulus G'

Viscous Modulus G''

Back to Gel point analysis

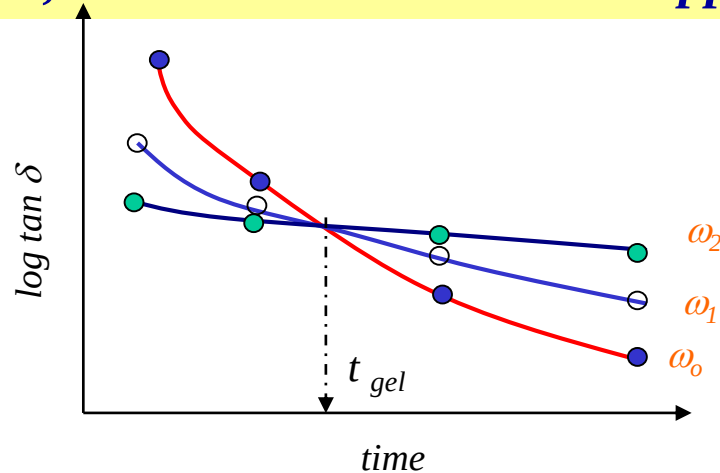
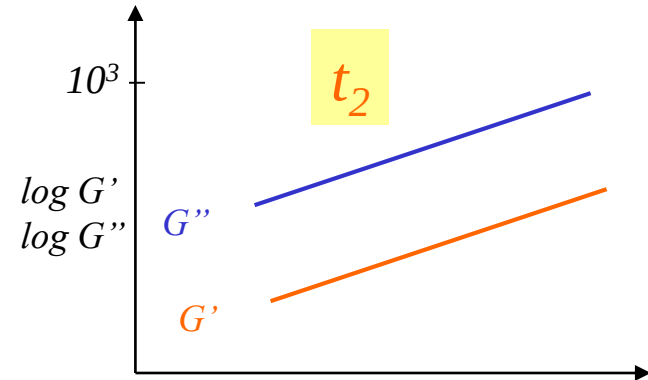
Concern:

*What if we have large enough time interval
so we miss gelation point (t_2 behavior)*

Solution:

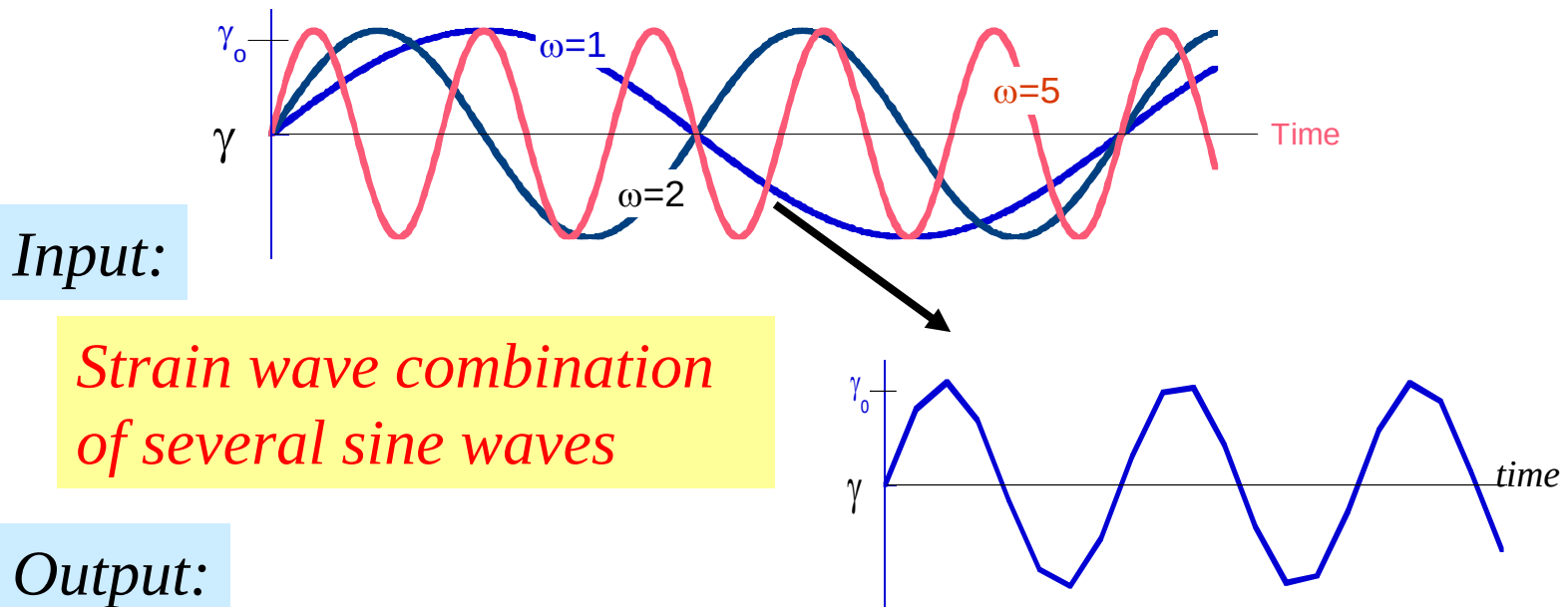
Plot $\tan \delta$ versus time

For fast reactions, in addition use Multiwave approach



Multiwave Approach

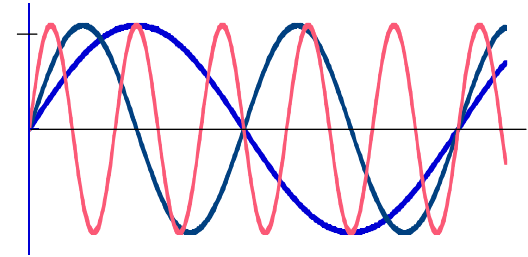
Combine several frequencies/oscillation into one.
Fourier Transform Mechanical Spectroscopy



Complex torque signal $\xrightarrow{\text{Fourier transform}}$

G', G'' for several frequencies from one data point

Multiwave Approach, cont'd



Advantages:

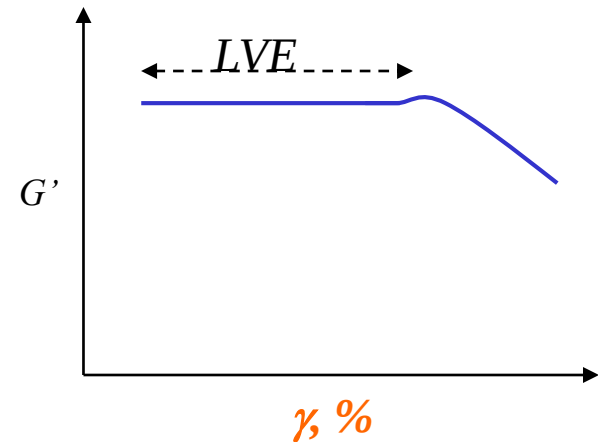
- *Effective in fast reacting system*
- *Only one data point gives several frequencies*
 - Can get G' , G'' versus ω from 1 data point*
- *Can monitor x-linking quickly*

Multiwave Approach, cont'd

Pick frequency ω_1 : other frequencies have to be multiples of ω_1

Input is superposition of multiple oscillatory strains: $\gamma = \sum_{i=1}^m \gamma_i \sin \omega_i t$

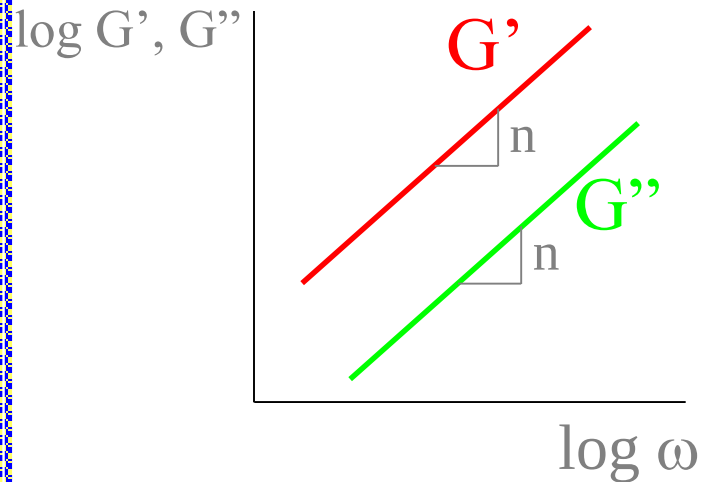
Linearity condition
 $\gamma_1 + \gamma_2 + \gamma_3 + \dots + \gamma_{10} < LVE \text{ limit}$



Criterion for Gel Point

Winter-Chambon Criterion

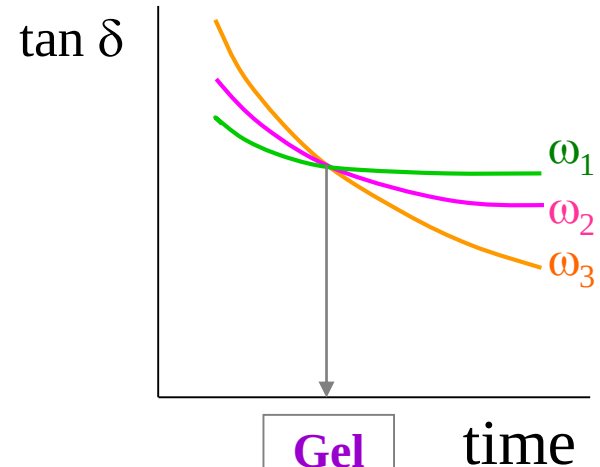
Frequency sweeps at gel point



Elastic Modulus $G' \sim \omega^n$

Viscous Modulus $G'' \sim \omega^n$

Loss tangents at different frequencies



Gel Point

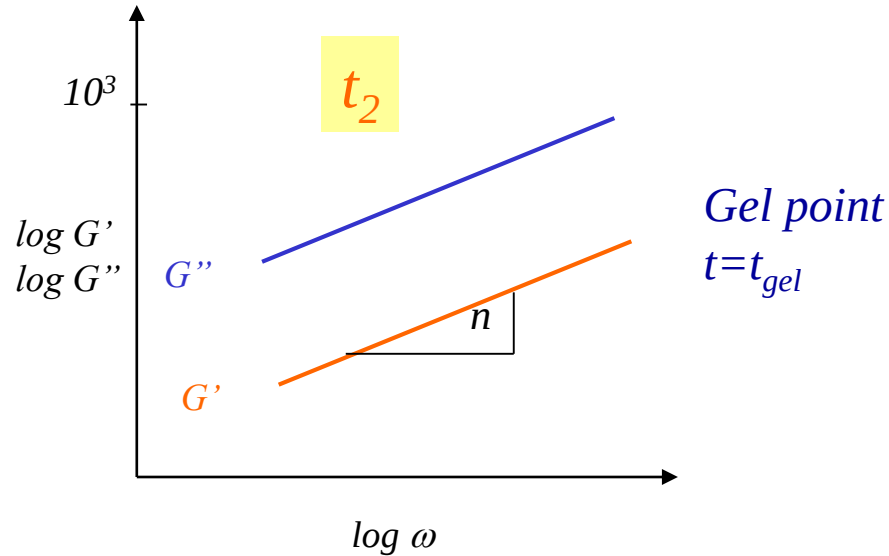
$$\tan \delta = \frac{G''}{G'} = \tan \left(\frac{n\pi}{2} \right)$$

(**independent** of frequency)

Gel & Network-Fractals

Concept of fractals:

Gel point data $\Rightarrow$ *Fractal Dimension* $\Rightarrow$ *Mesh size*



n is related to fractal dimension (discuss this more later)

Gel&Network-Fractals, cont'd

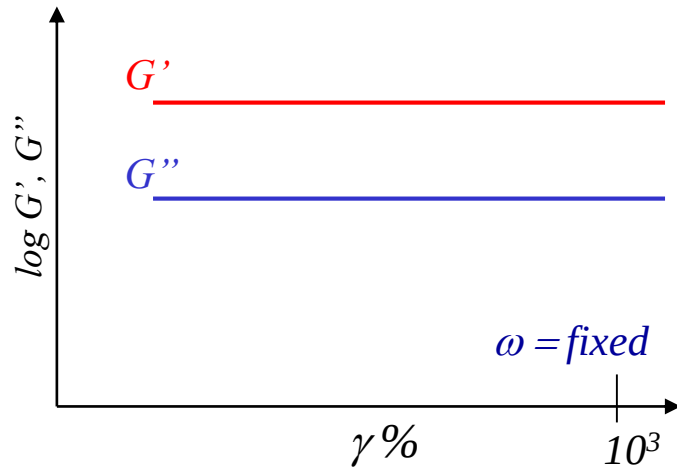
n is related to fractal dimension:

$$n = \frac{d(d + 2 - 2d_f)}{2(d + 2 - d_f)}$$

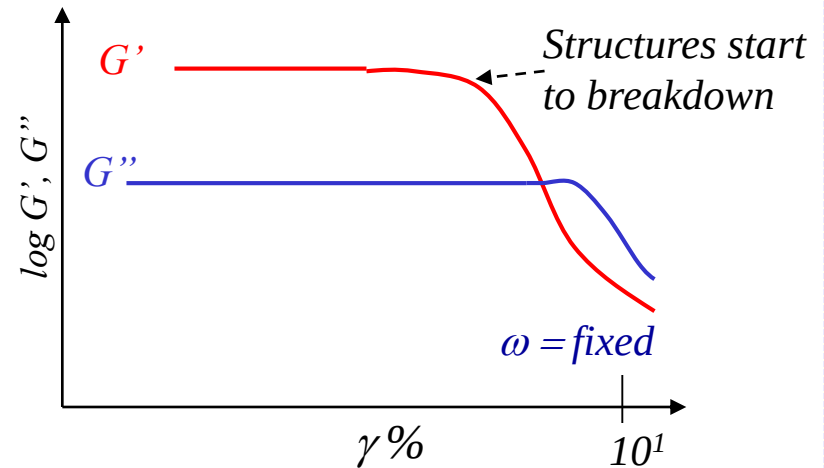
- *d_f is the fractal dimension (can be fractional)*
- *d is the dimension of the sample (=3)*

Physical versus Chemical Gel Systems

Chemically α -linked gels



Physical gels



- Physical gels may show **thixotropy**
- Can preshear sample and follow $G'(t)$ to observe structure recovery

Moving Forward

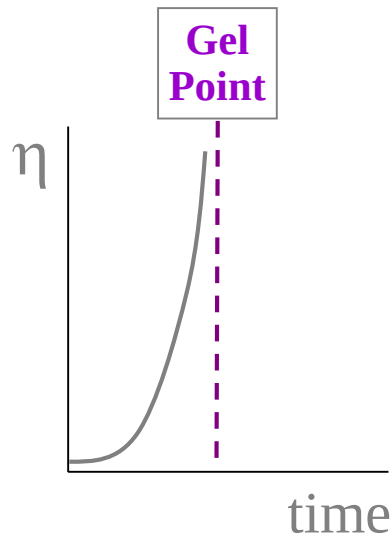
More on fractals – what is it?

Real data – gelation and gel point

Physical vs chemical gels

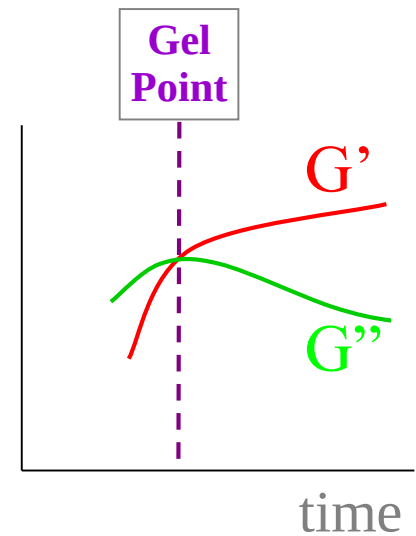
Thixotropy

Using Rheology to Detect Gel Point



Divergence of Viscosity

Intrusive measurement,
requires extrapolation



Dynamic Moduli $G'=G''$

Not a valid criterion
in general

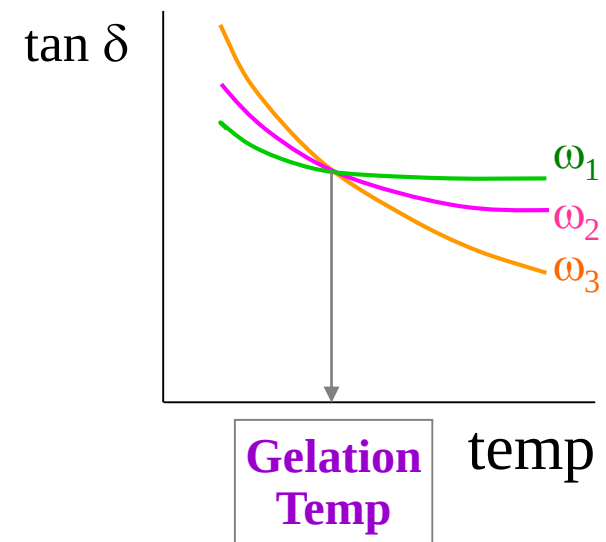
In-class problem

A PEO grafted PS latex system undergoes (thermoreversible) gelation when heated. How would you determine the temperature of gelation?

Do frequency sweeps of G' , G'' at different temperatures



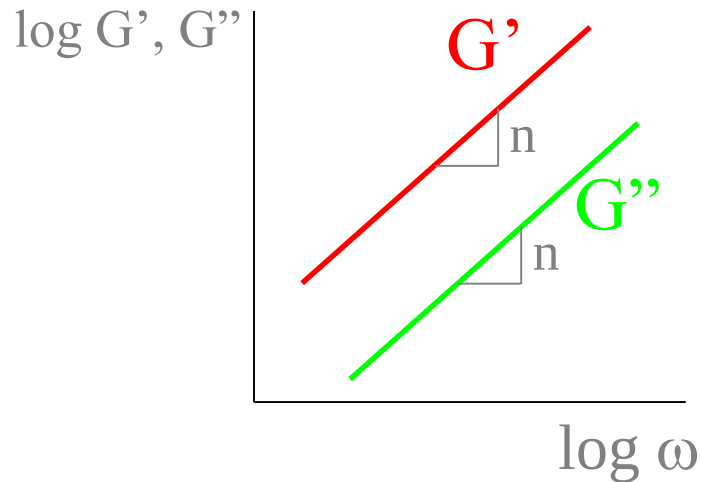
Pick $\tan \delta$ at specific frequencies (say ω_1 , then ω_2) and plot as function of temperature.



Loss tangents at different frequencies

Criterion for Gel Point

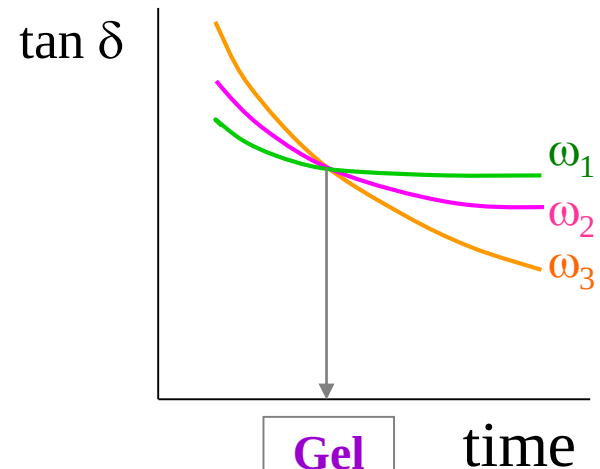
Frequency sweeps at gel point



Elastic Modulus $G' \sim \omega^n$

Viscous Modulus $G'' \sim \omega^n$

Loss tangents at different frequencies



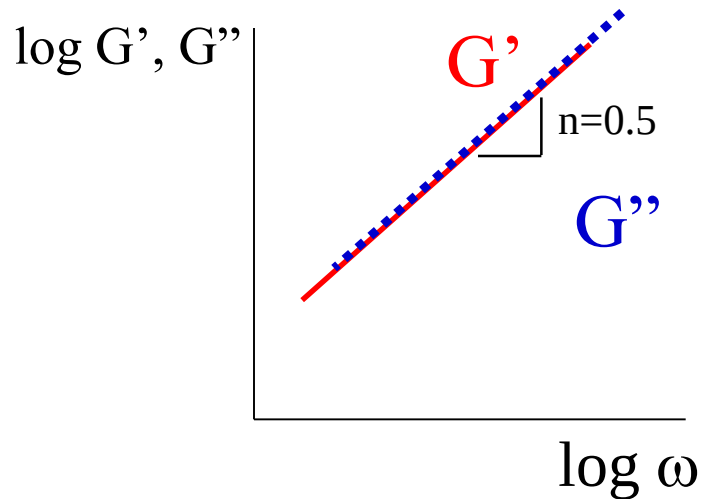
$$\tan \delta = \frac{G''}{G'} = \tan \left(\frac{n\pi}{2} \right)$$

(**independent** of frequency)

Gel Point Problem

Crosslinking of a polymer shows G' to scale as $G' \sim \omega^{0.5}$ at gel point

Plot G' and G'' in the graph below.



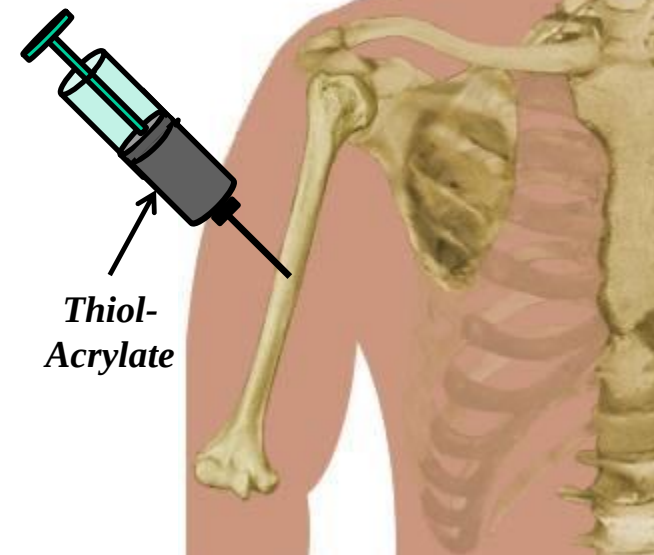
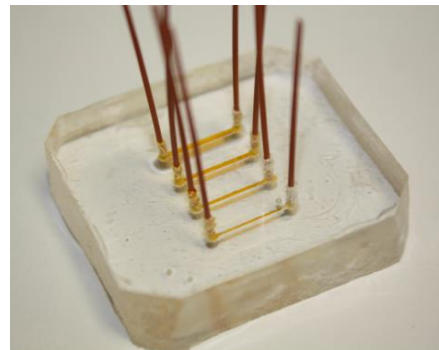
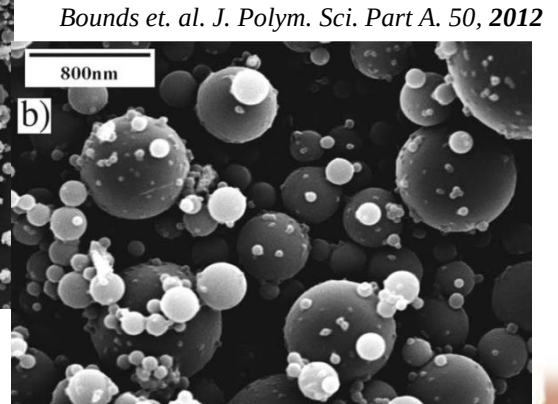
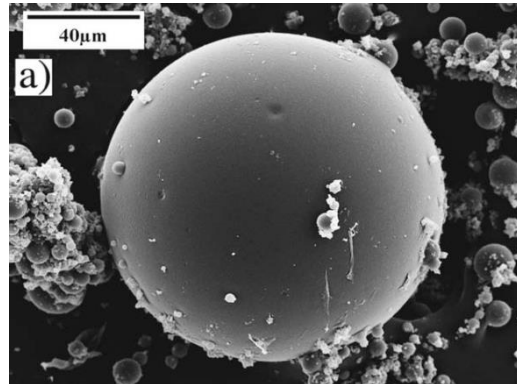
Elastic Modulus G'

Viscous Modulus G''

Rheological Monitoring of In situ Crosslinking

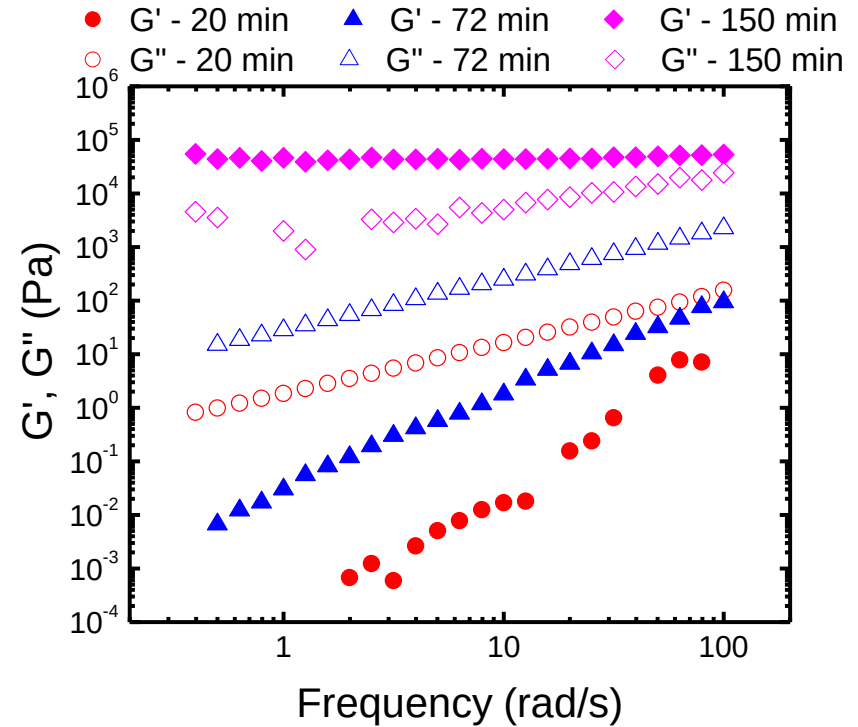
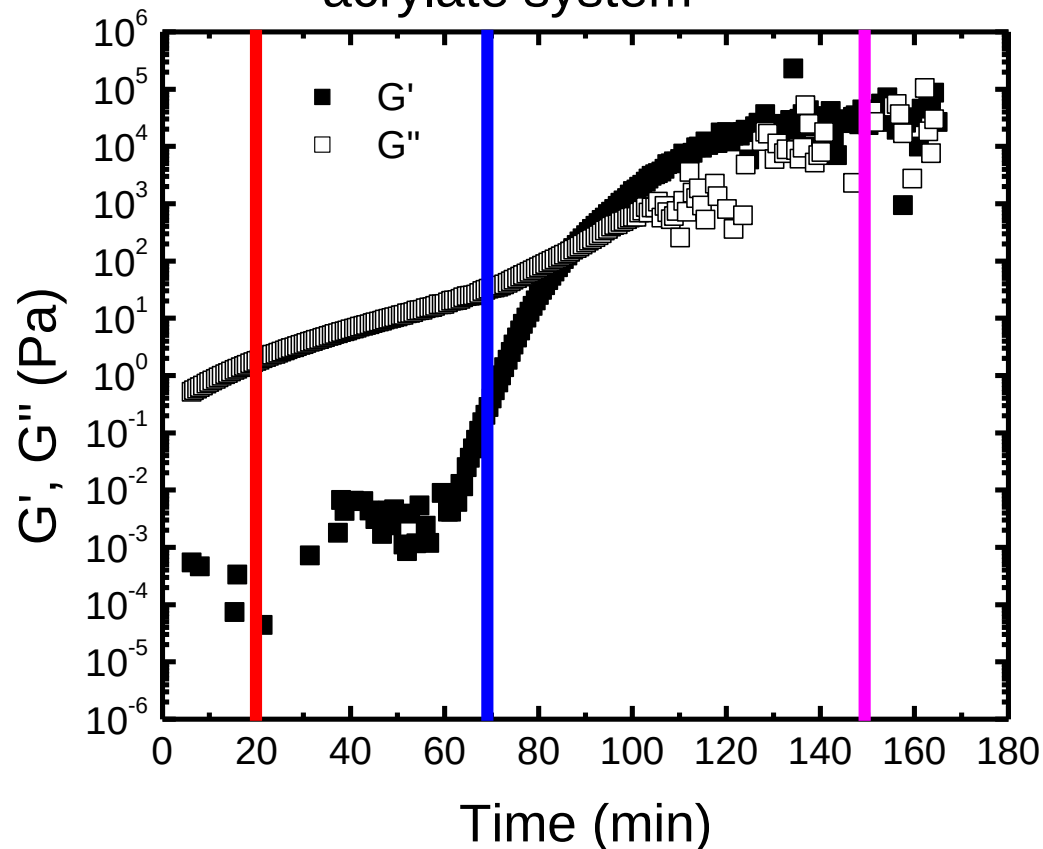
Polymerization of Thiol-Acrylate

- Benefits
 - Solvent free
 - Rapid synthesis
 - High conversions
 - Low shrinkage
 - Resistant to water absorption
- Applications
 - Microfluidics
 - Microparticles
 - Hydrogels
 - Surface modification
 - Biomaterials
- Often photopolymerized



Sol-Gel Rheology

Used FTMS approach to monitor moduli during crosslinking of a thiol acrylate system



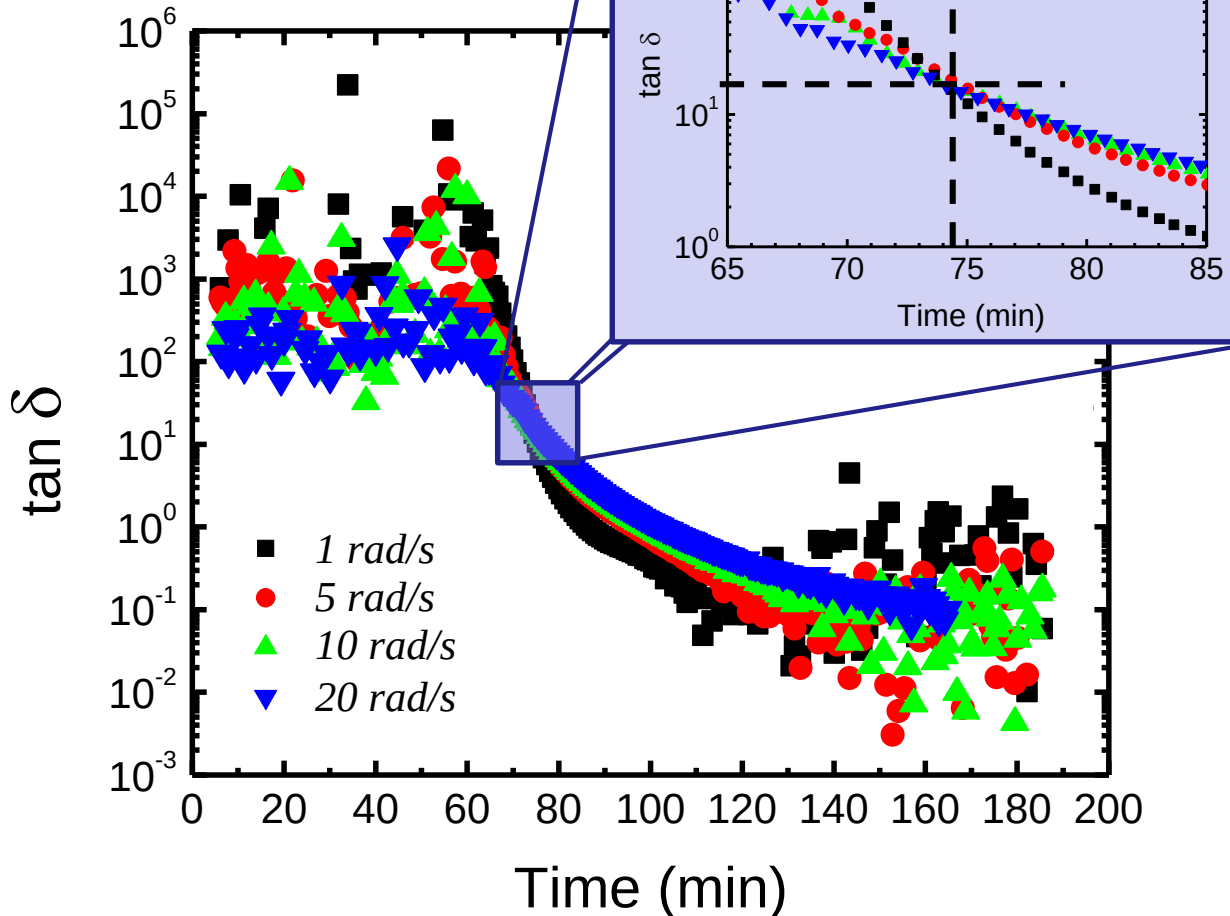
20 min: $G'' > G'$, pregel behavior

70 min: increased moduli

150 min: $G' > G''$ for all frequencies, crosslinked gel

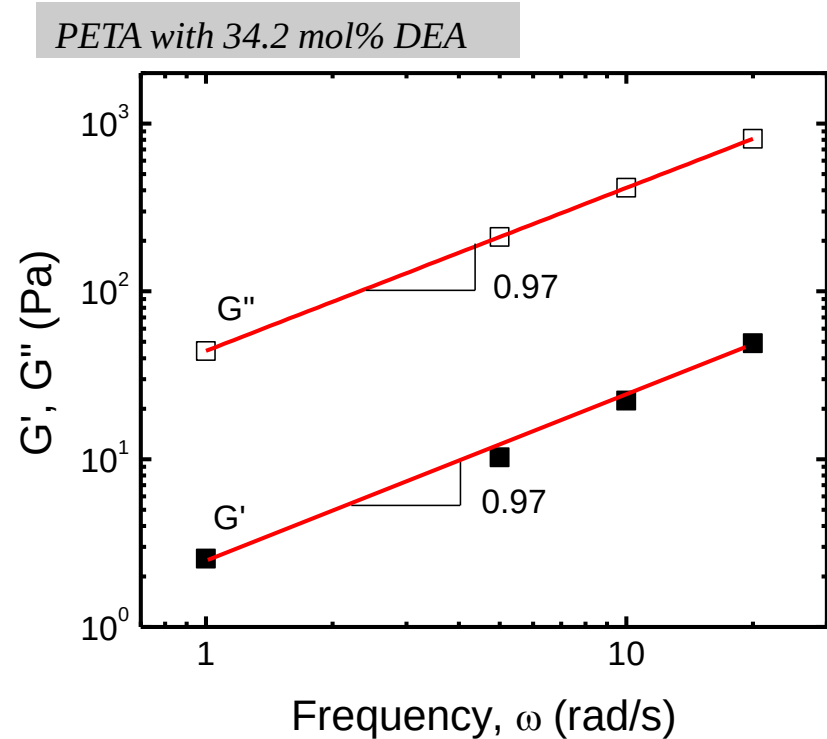
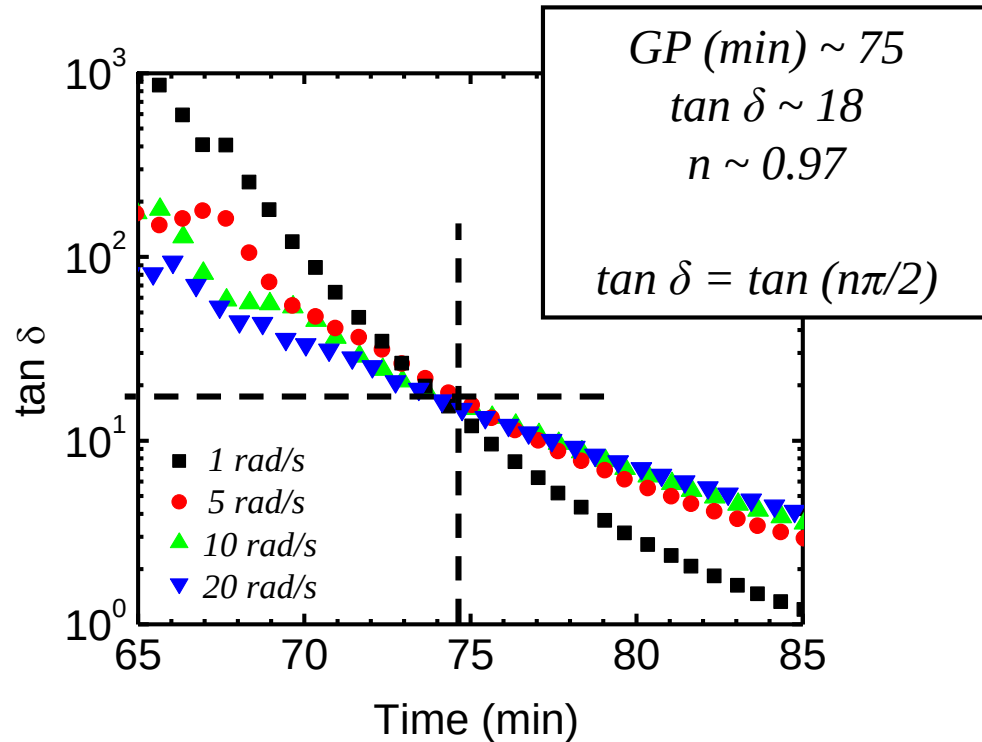
Gel Point Analysis

PETA with 34.2 mol% DEA



- All data obtained via FTMS approach, verified by time sweeps
- Gel point characteristic of system, thus $\tan \delta$ independent of frequency
- Time at $\tan \delta$ intersection represents gel point ($t \approx 75$ min)

Gel Point Analysis



- All data obtained via FTMS trials at $t \approx 75$ min (gel point)
- Using value of $\tan \delta$ at crossover we use: $\tan \delta = \tan (n\pi/2)$ to calculate theoretical slope of G' and G'' at gel point which correlates well with experiment
- Can use n to get fractal dimension of gel structure

Hydrogels

What is a hydrogel?

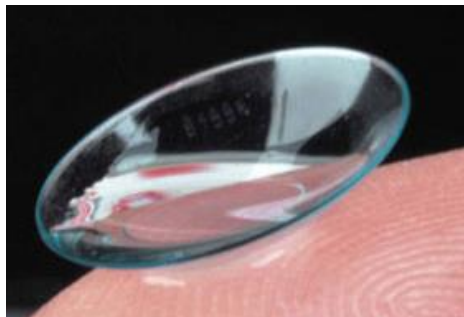
A network of polymer chains where water is the dispersion medium

Applications:

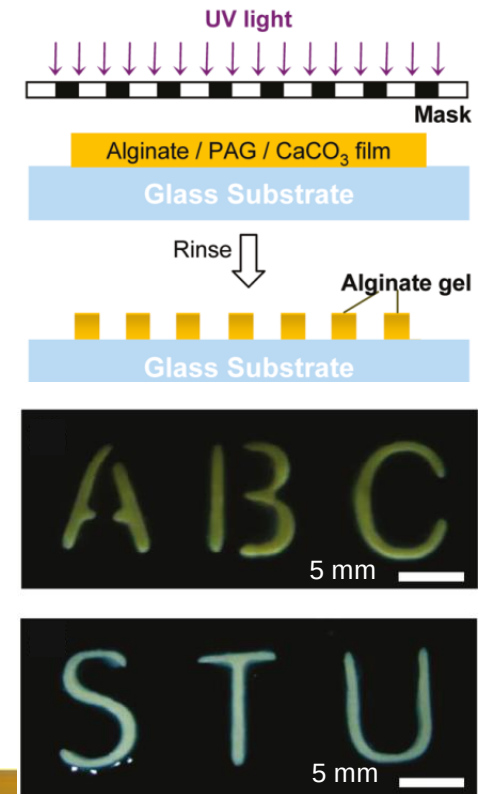
- Absorbents (99.9% water)
- Contact lenses
- Photopatterning
- Release of biomaterials (proteins, drugs, cells)
- Biosensors/Biocatalysis
- Tissue engineering



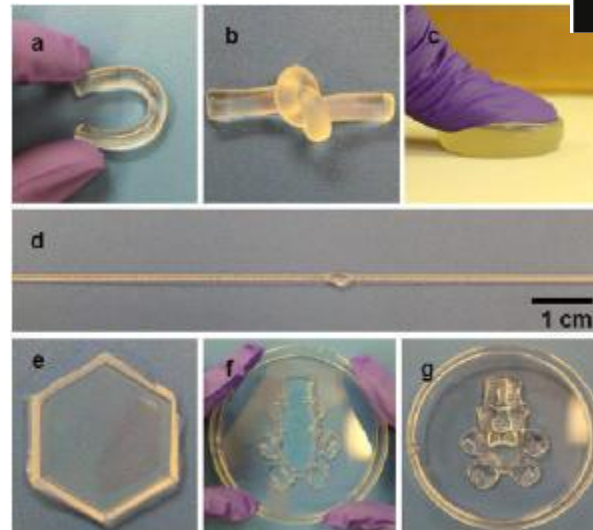
http://www.jacobsschool.ucsd.edu/uploads/news_release/2012/hydrogels_lr.jpg



http://maxim-eyes.co.uk/images/gas_permeable_contact_lens.jpg

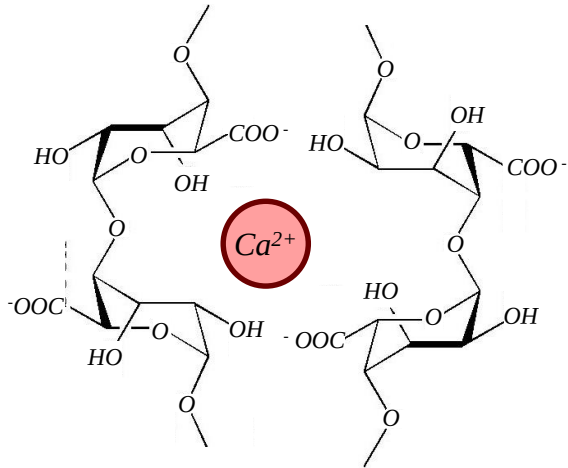


Javvaji et. al. *Langmuir* 27, 2011



<http://www.sciencedaily.com/releases/2013/07/130705212234.htm>

Photoinitiated Alginate Hydrogel



Before UV: Sol

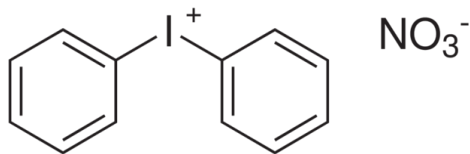


After UV: Gel

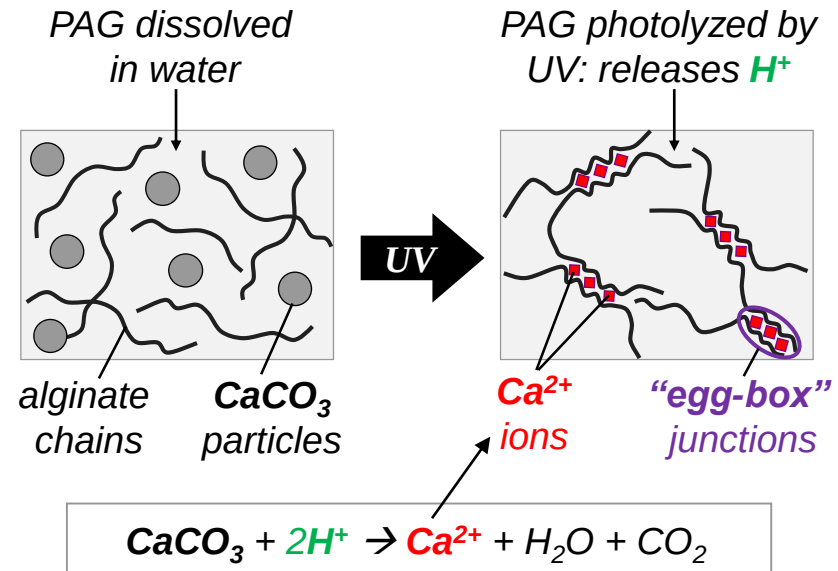


- **Sodium Alginate:**
 - 12 – 40 kDa MW
 - *Macrocystis pyrifera*
 - 18% G-block, 40% M-block, 42% MG-block

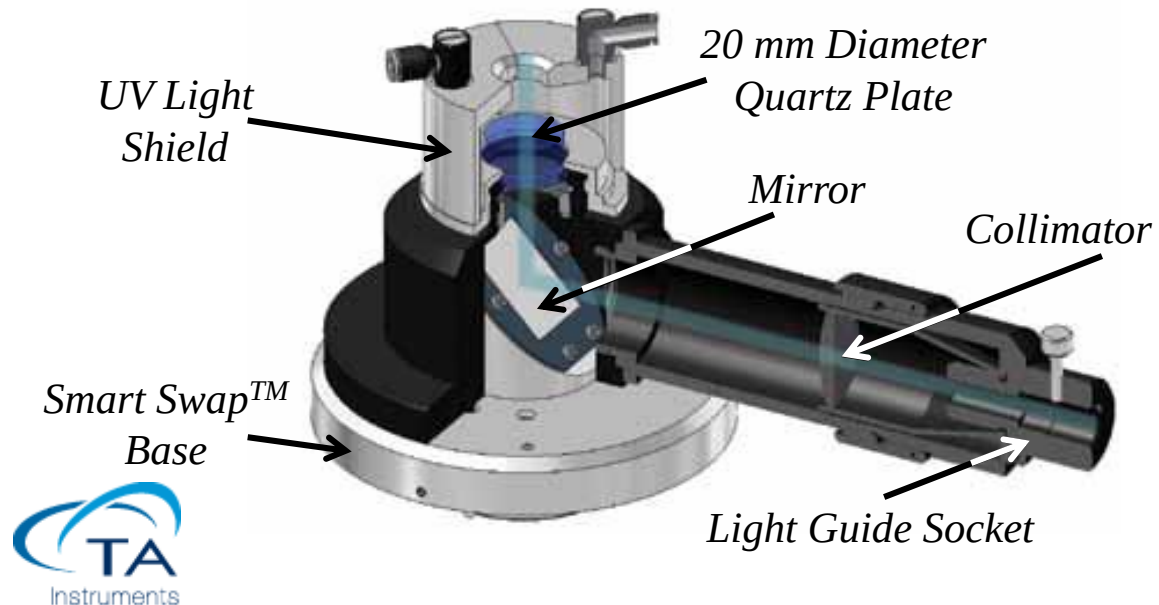
- **PAG:** Diphenyliodonium Nitrate



- **Ca²⁺:** Calcium carbonate particles
 - Diameter: ~70 nm



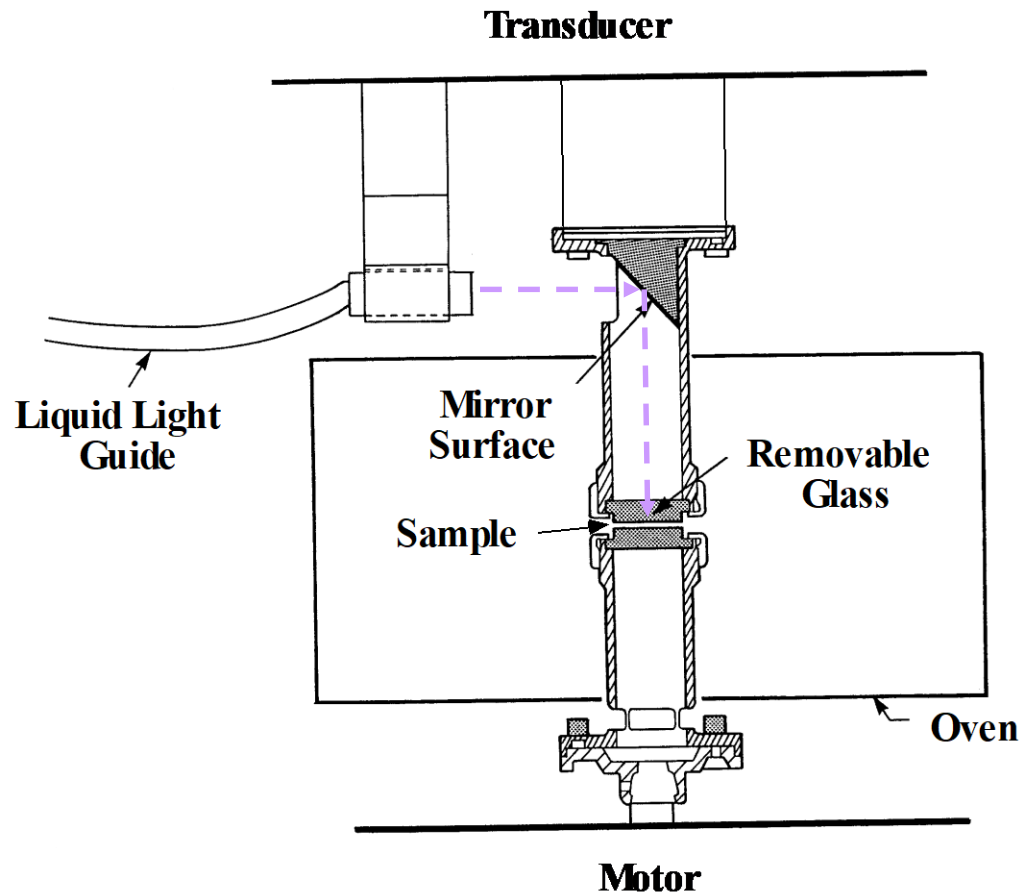
In Situ UV Rheology



- *External mercury lamp UV source*
- *Light guide and reflected mirror assembly*
- *250 nm – 600 nm, variable intensity (max: 300 mW/cm²)*
- *Uniform UV exposure to sample across quartz glass plate*

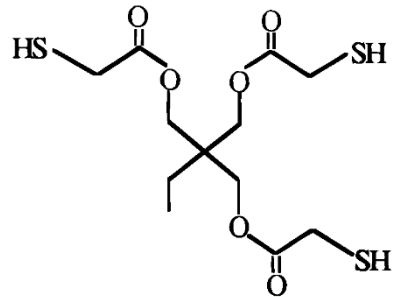
*How are hydrogel properties
and kinetics affected by:*
Solution Composition
UV Intensity
UV Exposure

In-situ Rheology during UV-Crosslinking

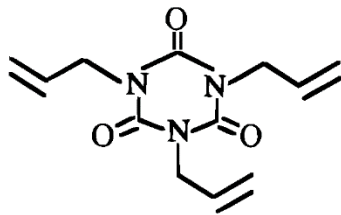


Apparatus designed for in-situ studies on RMS-800 rheometer

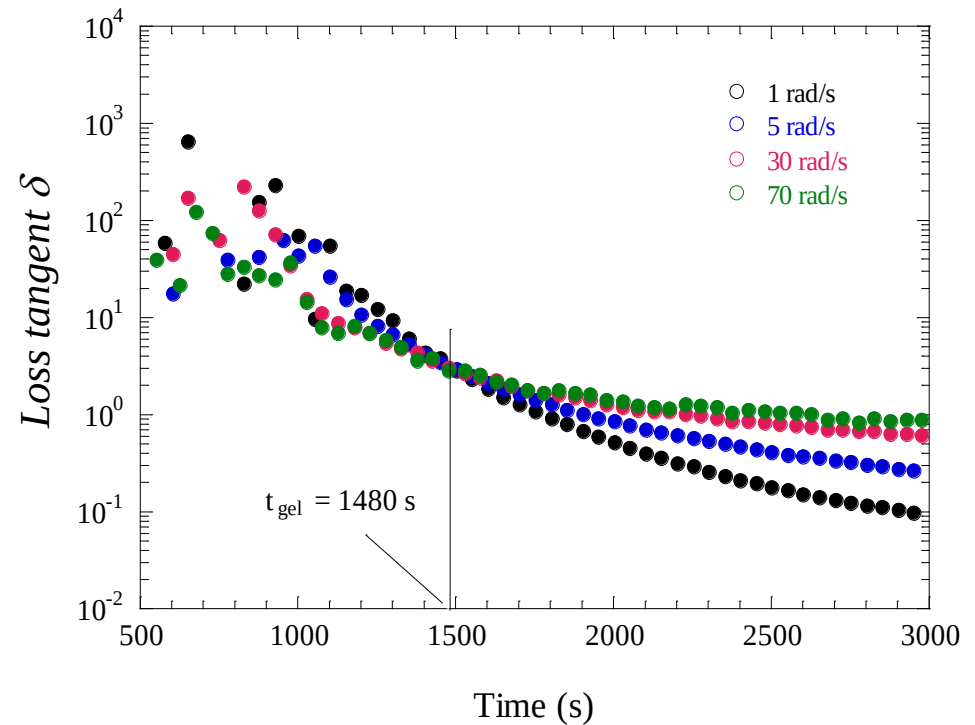
Model UV-Crosslinkable System: Thiol-ene



Trimethylolpropane tris(2-mercaptoacetate)



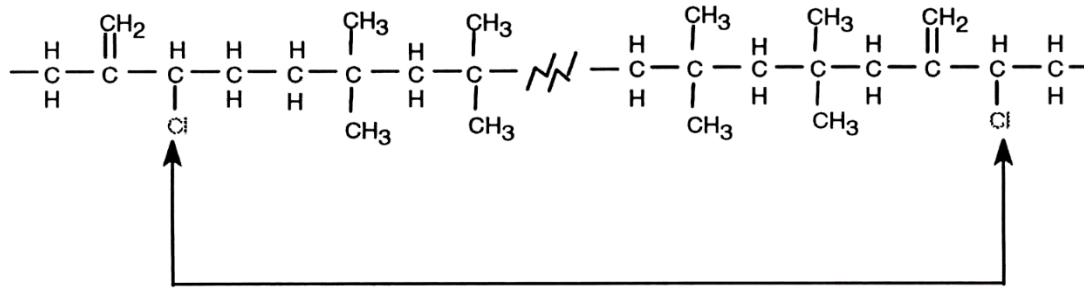
Triallyl Isocyanurate



Polymer obeys WC criterion with
a gel time of 1480 s

Studies with a Commercial Elastomer System

Is the Winter criterion applicable to the crosslinking of a commercial elastomer?



Chlorobutyl rubber

Polymer MW $\sim 500,000$

MW between crosslinks ~ 5000

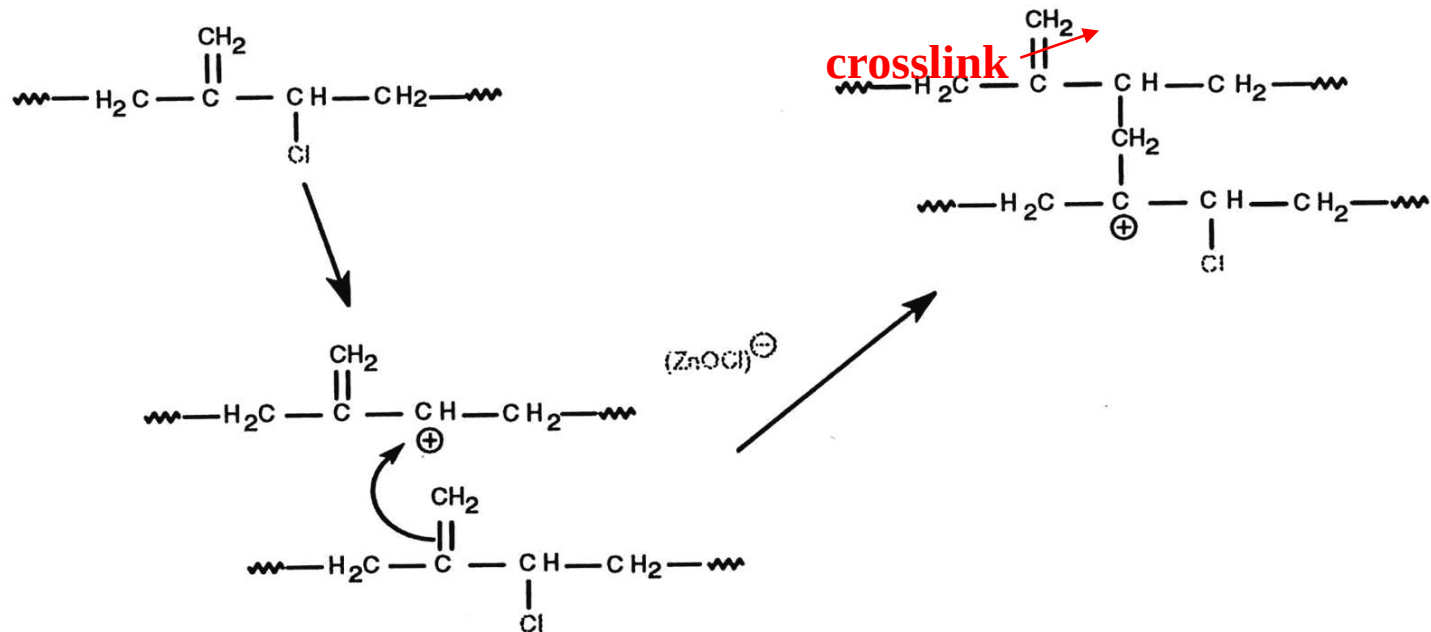
Contains silica, talc and clay fillers
(50 parts fillers per 100 parts polymer)

Samples obtained from Becton Dickinson Research Center as compounded stock.

Used for closure devices in biomedical applications due to its low permeability.

Crosslinking of Chlorobutyl Rubber

Crosslinking occurs at high temperatures ($T > 110^{\circ}\text{C}$)
in the presence of ZnO catalyst



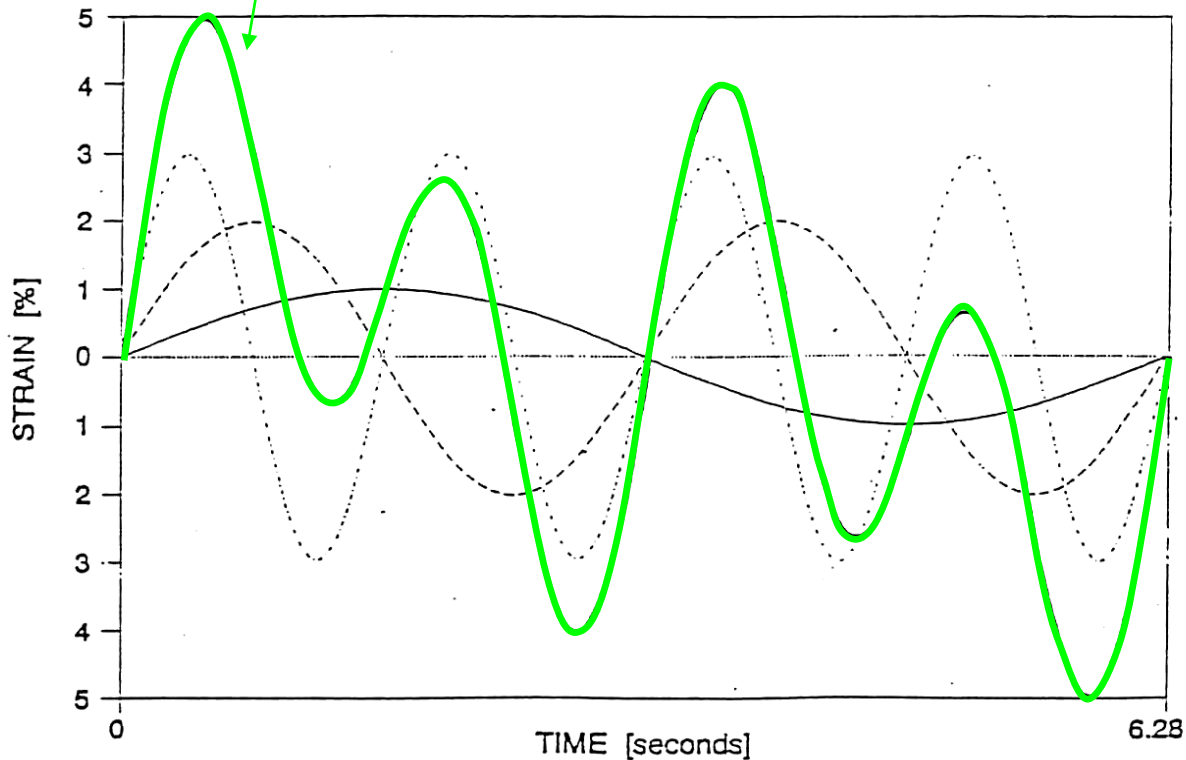
Multiple Waveform Rheology

Input is superposition of multiple oscillatory strains:

$$\gamma = \sum_{i=1}^m \gamma_i \sin \omega_i t$$

Linearity condition:

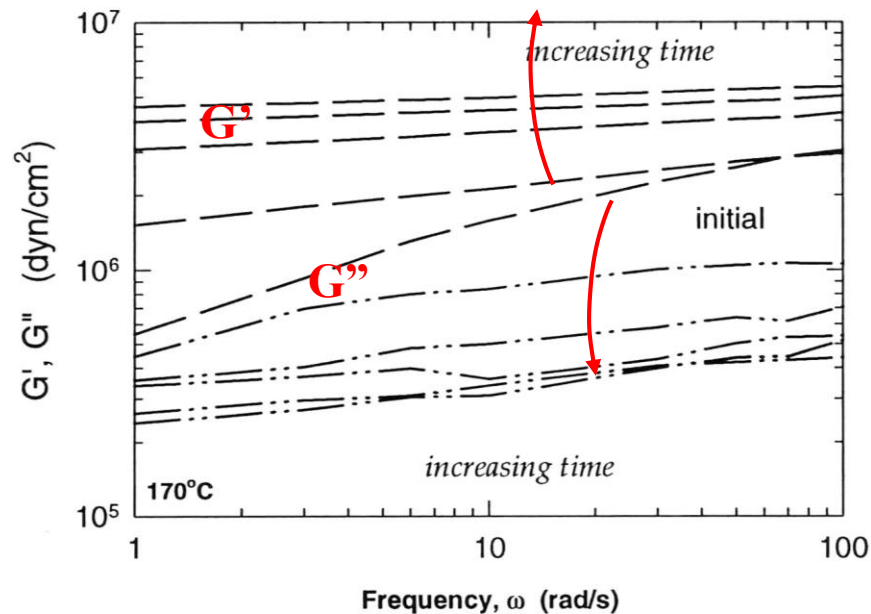
$$\sum_{i=1}^m \gamma_i \leq \gamma_c$$



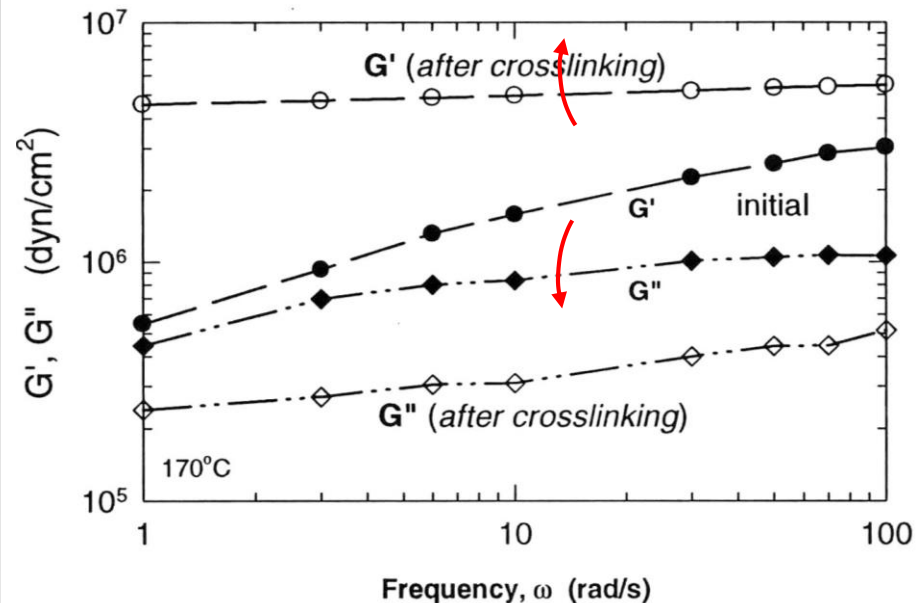
Advantage of speed: Simultaneous measurement of moduli (1-100 rad/s) in ~ 6 s

Multiwave Rheology of the Crosslinking Reaction

Data from a single multiwave experiment at 170°C



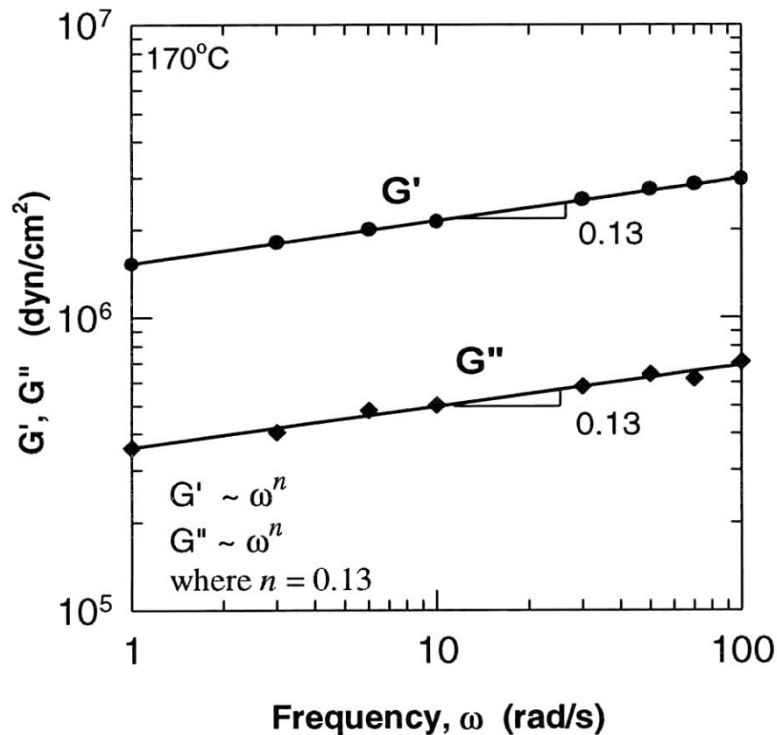
Frequency sweeps before and after crosslinking at 170°C



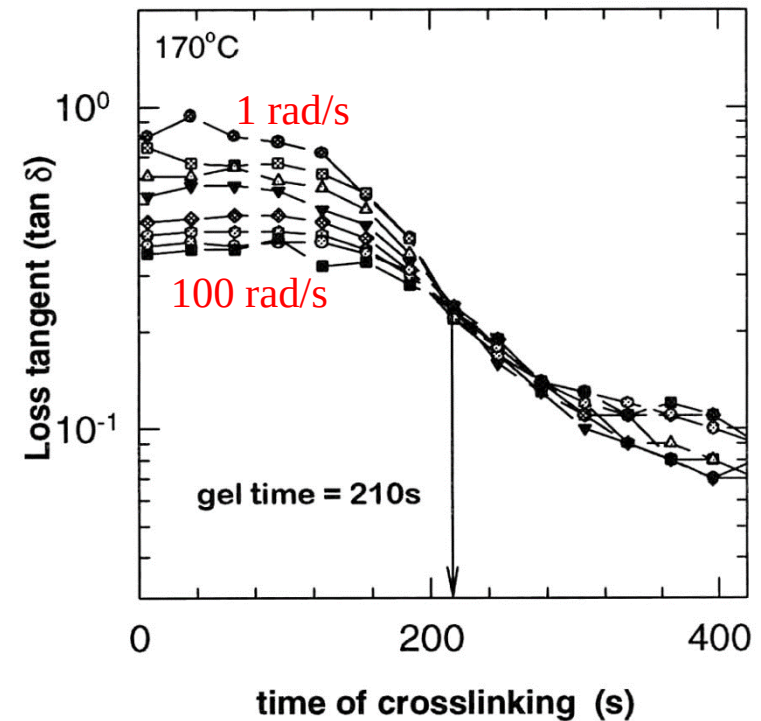
$G' > G''$ initially—does this make sense?

Gel Point Determination at 170°C

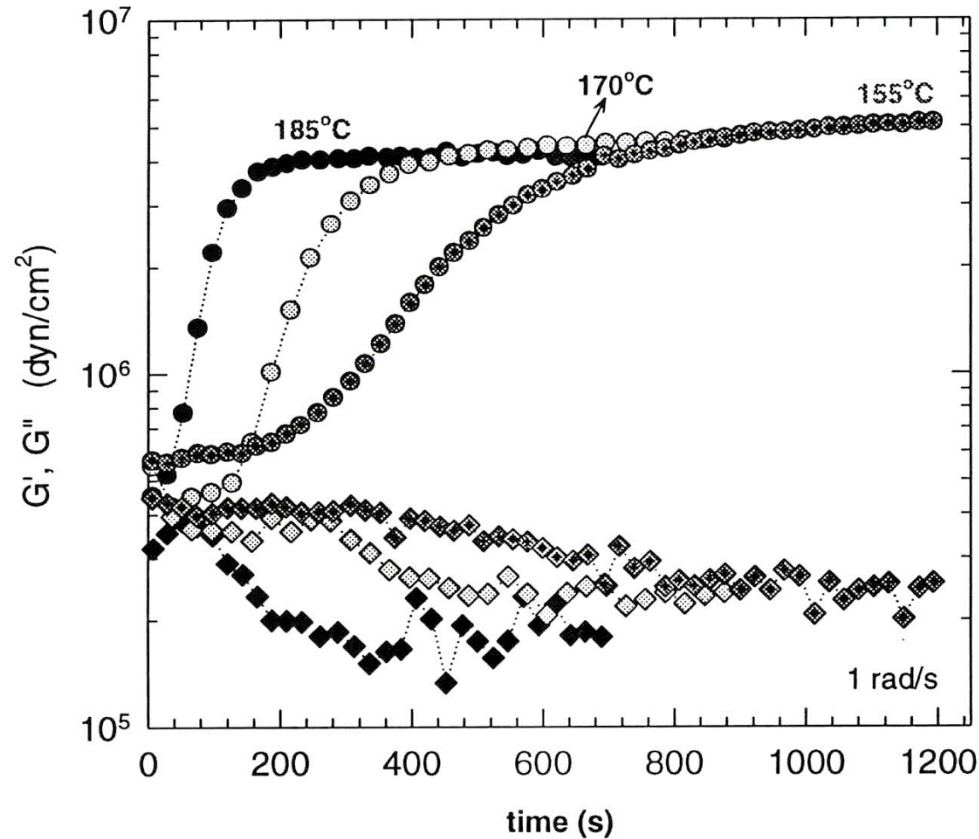
Frequency spectrum at $t = 210$ s



$\tan \delta$ vs. time at different frequencies

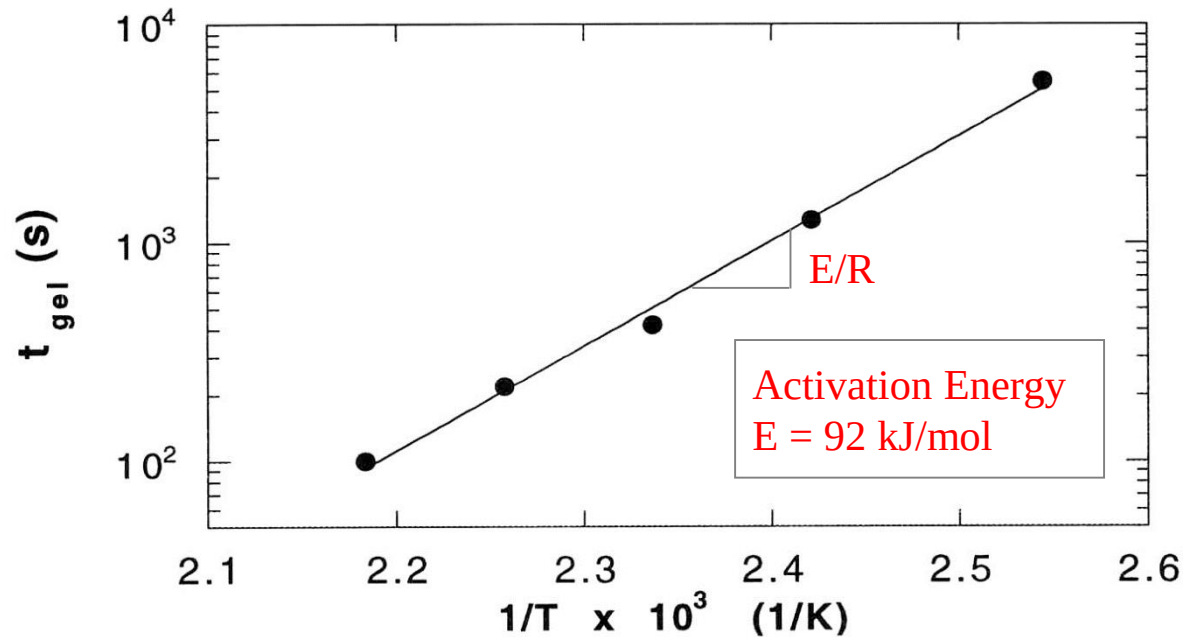


Crosslinking at Different Temperatures



Temperature of crosslinking (°C)	120	140	155	170	185
Gel time (s)	5470	1270	420	210	100

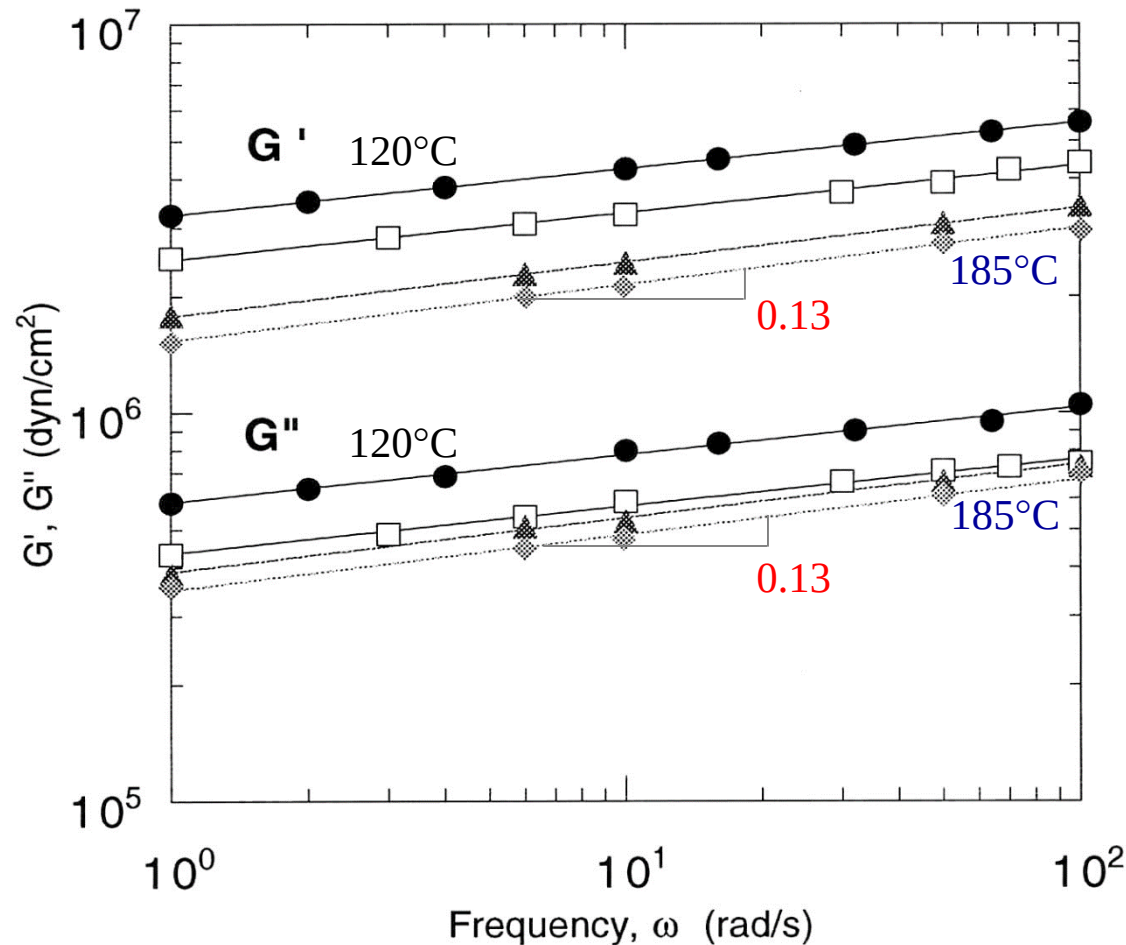
Activation Energy of Crosslinking Reaction



Assumptions

- Crosslinking is kinetically controlled until the gel point
- Conversion at gel point is independent of temperature

Critical Gel at Different Temperatures

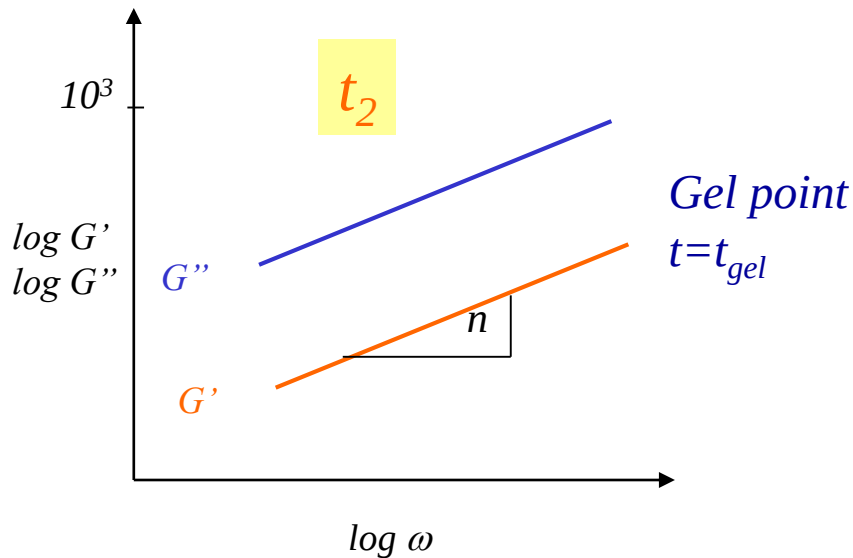


Relaxation exponent $n = 0.13$ independent of temperature

Gel & Network-Fractals

Concept of fractals:

Gel point data $\Rightarrow$ *Fractal Dimension* $\Rightarrow$ *Mesh size*

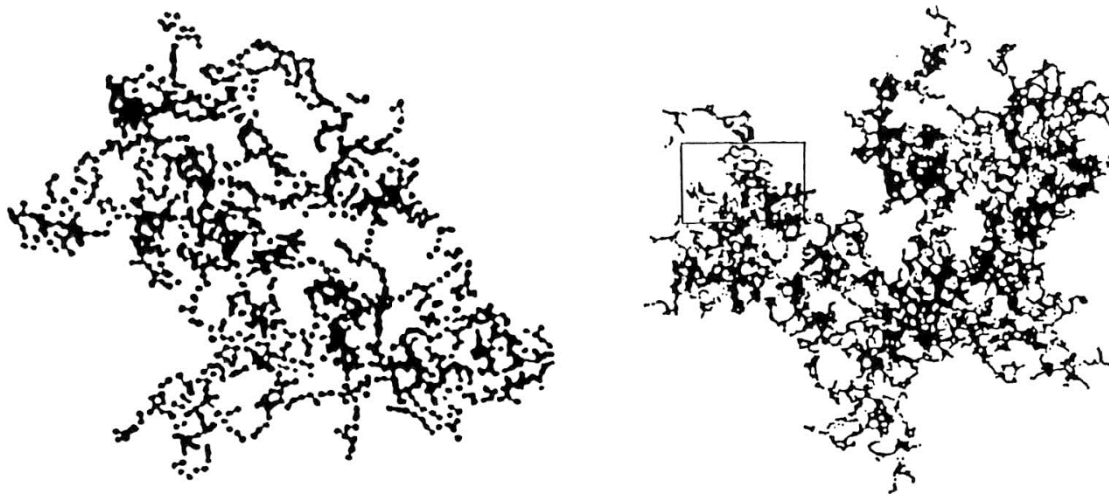


n is related to fractal dimension

$$n = \frac{d(d + 2 - 2d_f)}{2(d + 2 - d_f)}$$

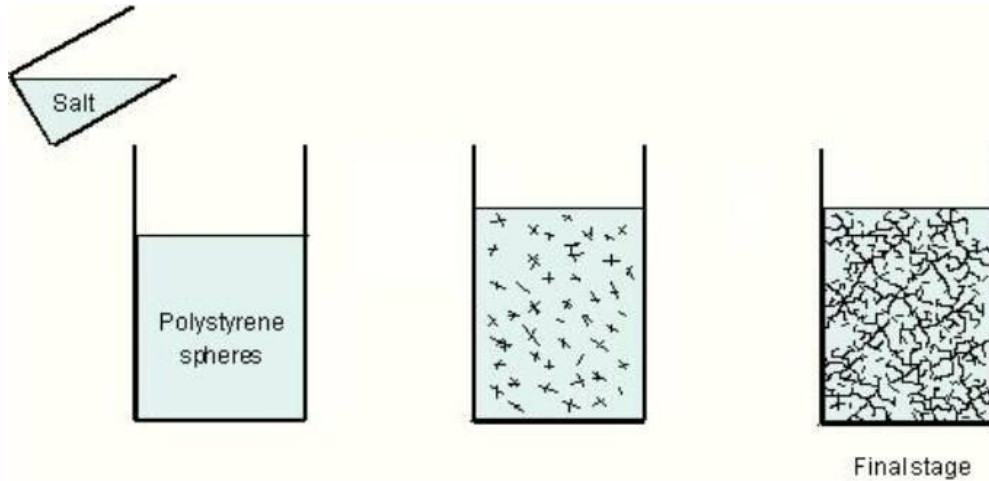
Lecture ended here

Self-Similarity of Critical Gel

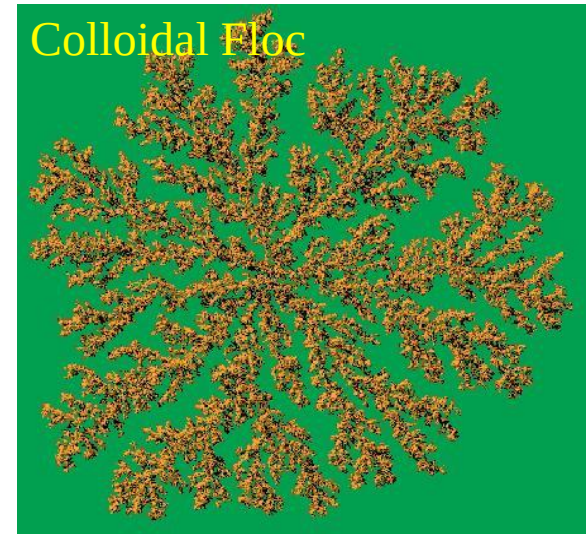


- At the gel point, the system is self-similar – a statistical fractal: $M \sim R^{d_f}$
- Fractal dimension d_f can be calculated from the value of n (Muthukumar, 1989)
- For $n = 0.13$, we get a fractal dimension $d_f = 2.38$
- High d_f indicates a dense fractal (due to high polymer MW, filler crosslinks)

Fractal Structure of Flocs and Gels

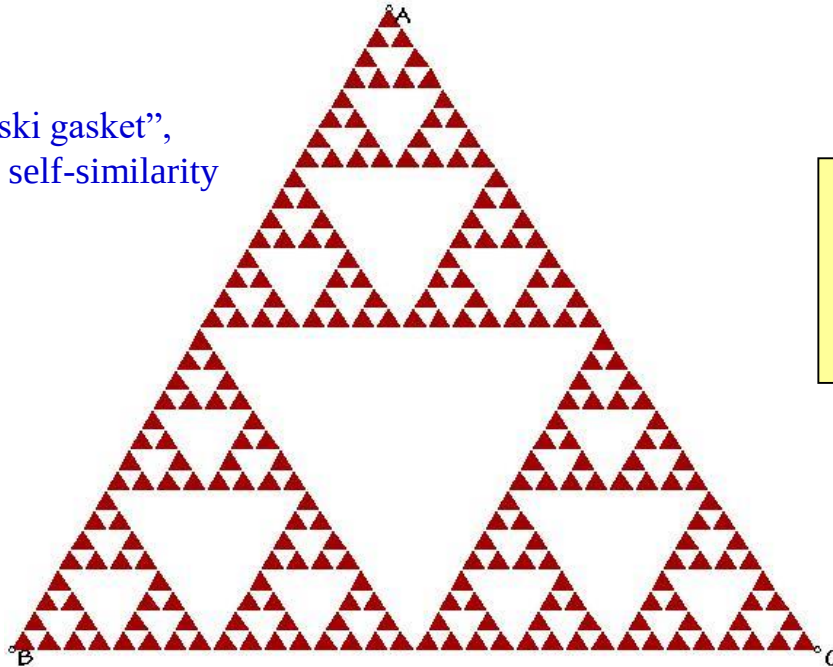


<http://www.deas.harvard.edu/projects/weitzlab/research/aging.html>



<http://www.ai.mit.edu/projects/im/broch/den1.html>

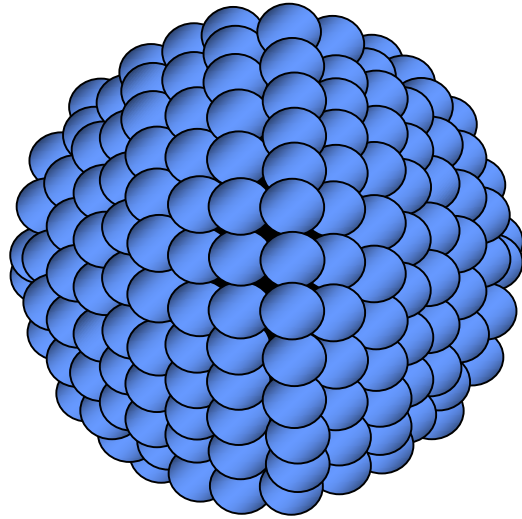
“Sierpinski gasket”,
showing self-similarity



<http://www.math.uncc.edu/~droyster/>

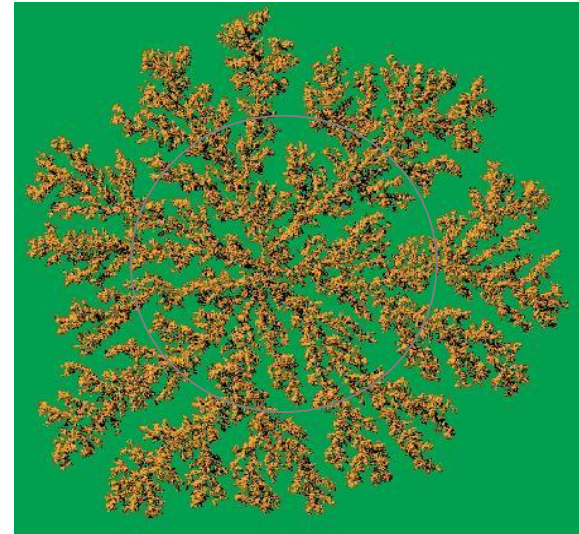
Colloidal flocs and gels tend to have a loose, open structure. They show the property of scale invariance or self-similarity and are hence referred to as fractals.

Fractal vs. Non-Fractal Clusters



Non-Fractal Spherical Cluster

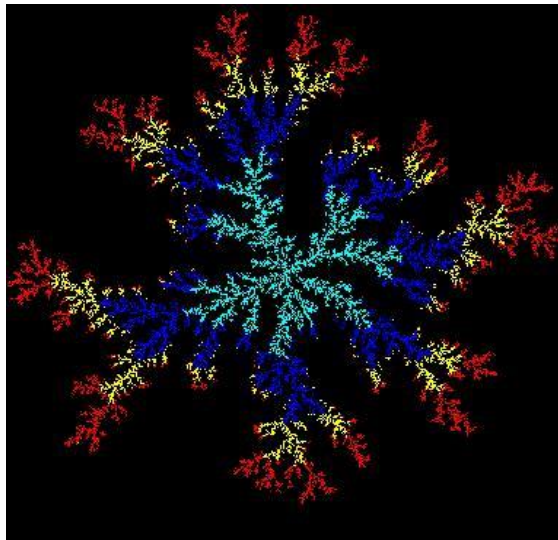
- The mass $M \sim R^3$
- Density is independent of size: $\frac{M}{V} \sim R^0$



Fractal Cluster

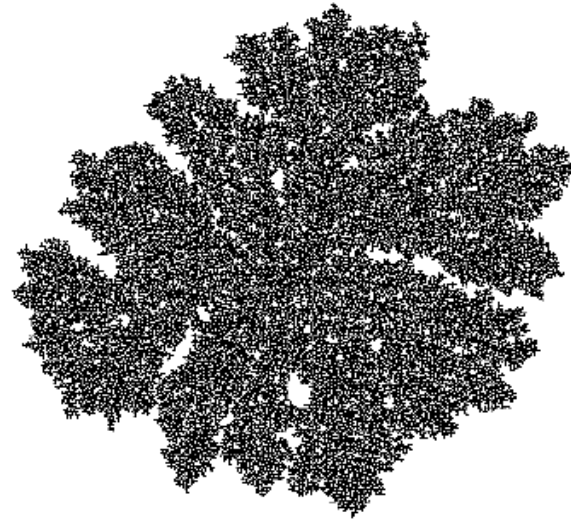
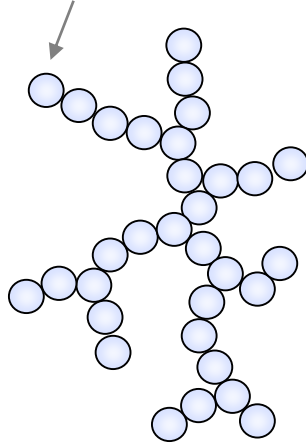
- The mass $M \sim R^{d_f}$
- Density **decreases** with size: $\frac{M}{V} \sim R^{d_f-3}$
- The fractal dimension d_f is a number < 3 .
- The lower the d_f , the **more open** the structure

Fractal Clusters and Aggregation Mechanisms



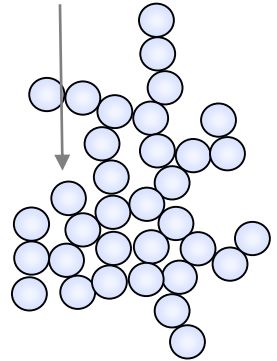
<http://www-personal.umich.edu/~lsander/>

incoming particle
sticks to the
periphery of the floc



<http://www.physics.uc.edu/~pinski/dla/dlapres/>

incoming particle
bounces around before
finding a destination



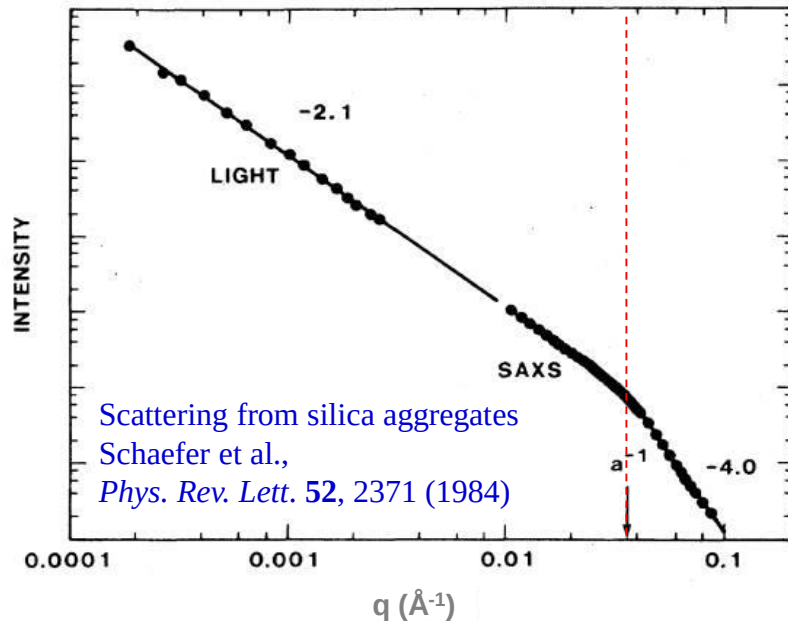
Diffusion-Limited Aggregation (DLA)

- Particles are strongly attracted to each other
- High “sticking probability”: stick on contact
- Diffusion through the fluid is the limiting step
- Leads to open flocs ($d_f = 1.75$)

Reaction-Limited Aggregation (RLA)

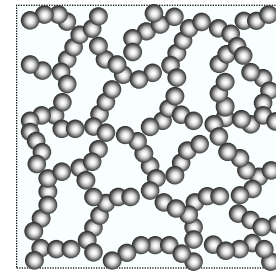
- Particles are weakly attracted to each other
- Low “sticking probability”
- Reaction is the limiting step
- Leads to more compact flocs ($d_f = 2.0 - 2.2$)

Fractal Clusters: Scattering and Rheology

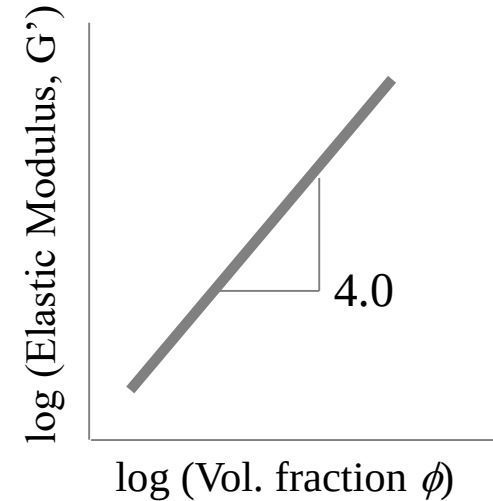


Scattering from Fractal Dispersions

- Intensity I varies as a power law: $I \sim q^{-d_f}$
- Slope gives the fractal dimension d_f
- Here $d_f = 2.1 \Rightarrow$ RLA



GEL



Rheology of Fractal Gels

- Gel $\Rightarrow$ 3-D network: G' indep. of frequency
- As part. vol. fraction $\uparrow$, $G' \uparrow$ in power law
- $G' \sim \phi^{4.0-4.5}$ for DLA ($d_f = 1.75$)

Food for thought

Loss tangent values at gel point

G' , G'' vs frequency plot shapes for gels

Relaxation modulus G at gel point

Implication of the flatness of G' for gels

Relaxation time of a gel